3 Numerical methods and comparison protocol

Section 2 showed that A is sparse, symmetric and positive definite, with sensitivity measured by $\kappa_2(A)$. These properties decide which solvers apply and how fast they converge. Let $N = |F|$ be the number of unknowns, and x_k, $e_k = x - x_k$ and $r_k = b - Ax_k$ the k-th iterate, error and residual.

A direct method factorizes A as $PA = LU$ or, for symmetric positive definite A, as $A = LL^T$ (Cholesky) at half the cost, then solves two triangular systems [7]; forming A^{-1} costs more and is less accurate. Iterative methods avoid the extra nonzeros (fill-in) of sparse factors and only multiply by A.

A stationary iteration corrects x_k with the residual, using an easily inverted approximation M of A: $x_{k+1} = x_k + M^{-1}r_k$, so $e_{k+1} = (I - M^{-1}A)\,e_k$. It therefore converges for every x_0 if and only if the spectral radius $\rho(I - M^{-1}A)$ is below one [3]. Jacobi uses $M = D_A$, the diagonal of A. Its iteration matrix is similar to $I - S$ with $S = D_A^{-1/2} A D_A^{-1/2}$, so

$$\rho_J = \max_i |1 - \lambda_i(S)|. \tag{10}$$

As $\lambda_i(S) > 0$, Jacobi converges if and only if $\lambda_{\max}(S) < 2$. Damped Jacobi, $M = D_A/\omega$, converges if and only if $0 < \omega < 2/\lambda_{\max}(S)$; we use $\omega = 0.5$. Gauss–Seidel takes M as the lower triangle of A and converges for every symmetric positive definite A [3].

The remaining methods exploit that x minimizes $\phi(y) = \frac{1}{2} y^T A y - b^T y$, the elastic energy of Eq. (8) minus the work of the loads. Steepest descent minimizes ϕ along $r_k = -\nabla\phi(x_k)$; each step multiplies the energy norm of the error, $\|e\|_A = (e^T A e)^{1/2}$, by at most $(\kappa - 1)/(\kappa + 1)$, where $\kappa = \kappa_2(A)$. The conjugate gradient (CG) method instead minimizes $\|e_k\|_A$ over $x_0 + \mathcal{K}_k$, where $\mathcal{K}_k = \operatorname{span}\{r_0, Ar_0, \ldots, A^{k-1}r_0\}$ is the Krylov space. It needs one product by A per step and satisfies [6]

$$\|e_k\|_A \le 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|e_0\|_A, \tag{11}$$

so its iteration count grows like $\sqrt{\kappa}$ rather than κ. Diagonally preconditioned CG (PCG) is equivalent to CG applied to S, so $\kappa_2(S)$ replaces κ; it helps only if the diagonal of A varies strongly. MINRES minimizes $\|r_k\|_2$ over the same space [1]. Full GMRES does so for any A but stores all k basis vectors [2], so it shows the cost of ignoring symmetry.

Comparison protocol. Every method receives the same A and b, starts from $x_0 = 0$ and is accepted only if its recomputed relative residual satisfies

$$\eta(\hat{x}) = \frac{\|b - A\hat{x}\|_2}{\|b\|_2} < 10^{-8}. \tag{12}$$

Reaching the cap of 20,000 iterations, or $\eta > 10^{10}$ for ordinary Jacobi, counts as failure. Accuracy is the relative error $\|\hat{x} - x_{\mathrm{ref}}\|_2 / \|x_{\mathrm{ref}}\|_2$ against the sparse LU solution x_{ref} of the same system [5]. Times are the best of three runs, excluding assembly and diagnostics. Extreme eigenvalues, for $\kappa_2(A)$ and ρ_J, come from dense or, for large systems, sparse eigensolvers [4].

Since $\hat{x}$ solves exactly the system with load perturbed by its residual, the standard bound [7]

$$\frac{\|\delta x\|_2}{\|x\|_2} \le \kappa_2(A) \frac{\|\delta b\|_2}{\|b\|_2} \tag{13}$$

limits its relative error to $\kappa_2(A)\,\eta$. This *algebraic* error measures how well one mesh's equations are solved. *Discretization* error, between finite element and continuum displacements, does not shrink with the tolerance; we estimate it by comparing meshes at matched physical points.

Later sections also check that d_C equals the prescribed values, that applied loads and the reactions of Eq. (7) have zero resultant force and moment, and that the response to a random

load perturbation obeys Eq. (13). A patch test pulls a plate held only by rollers (left edge fixed horizontally, corner at the origin vertically) with uniform traction p. Its exact solution $u_x = px/E$, $u_y = -\nu py/E$ is linear, so linear triangles must reproduce it to rounding on any mesh, verifying assembly, loading and reduction.

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